Prakhar Praveen

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WORK EXPERIENCE

Software Development Engineer, Seller Partner Services **Aug 2022 – Nov 2024**
Amazon Inc. **Tempe/Scottsdale, AZ**

- Architected a high-scale rule-authoring platform utilized by **200+ teams** across Amazon to streamline Fraud Prevention operations.
- Engineered a specialized Java library for platform access that slashed average monthly client latency by **25 hours**.
- Optimized system efficiency to cut server-side execution by **85%**, reducing annual infrastructure costs by **$30,000** through strategic operational caching.
- Automated the deployment lifecycle by transitioning to full CD, integrating component and end-to-end tests to save **260 developer-days** annually.
- Pioneered a local testing environment that accelerated development velocity by **2 days per month** while reducing AWS CloudFormation hosting costs by **$500 monthly** per account.
- Designed and delivered event-driven microservices using Java/Spring Boot on AWS to power critical seller-facing operations.
- Improved system scalability and performance by integrating **AWS services** such as **EC2, S3, Lambda**, and **DynamoDB**.
- Enhanced system observability by integrating distributed tracing and structured logging, significantly reducing MTTR and enabling proactive incident detection.
- Spearheaded DevOps best practices, including the "test pyramid" and staging canaries, to ensure safe and frequent production deployments.
- Utilized **AWS CloudWatch Logs e**xtensively for debugging and monitoring applications, leading to faster issue resolution and improved system uptime.
- Managed **AWS** resources and implemented **CI/CD pipelines** using **Jenkins** and **GitHub** Actions for continuous integration and deployment.

Graduate Teaching Assistant **Aug 2021 – May 2022**
Arizona State University **Tempe, AZ**

- Conducted Java programming lab sessions for undergraduate students, fostering collaborative learning and incremental development.
- Provided individualized support during office hours, assisting students with debugging and problem-solving.
- Assisted in grading assignments and exams, and contributed to curriculum development.

PROJECTS

Psychometric Intake Engine (prototype)

- Adaptive, diagnosis-aware screening (BPD/OCPD focus) with item-banked questionnaires, branching logic, scoring/normalization, clinician summaries, and patient-friendly heatmaps. Built on AWS Lambda + DynamoDB with tracing and safety guardrails.

Statistical Machine Learning: Movie Recommender System

- Built a robust recommendation engine utilizing unsupervised learning algorithms to predict user preferences.
- Evaluated and benchmarked the performance metrics of diverse recommendation strategies to identify the most accurate predictive model.
- Deployed machine learning models using Scikit-learn and TensorFlow to handle large-scale data processing and inference.

Natural Language Processing: Language & POS Tag Models

- Engineered high-accuracy Hidden Markov Models (HMM) in Python to automate language detection and part-of-speech tagging.
- Architected a Bayesian inference model to perform speaker attribution for the 2020 Presidential Debate using labelled linguistic datasets.

- Leveraged NumPy, SciPy, Pandas, and SpaCy to execute complex data preprocessing and model implementation pipelines.
- **Software Security CTF: Hacker**
- Spearheaded a competitive security team to a Top 10 finish out of 38 teams in a high-stakes Capture the Flag (CTF) tournament.
- Developed automated exploit scripts utilizing Python, C, and SQL injection techniques to identify and neutralize system vulnerabilities.
- Hardened containerized software environments by identifying and patching critical security flaws to pre-emptively block opponent exploits.

SKILLS

Languages:	Java, Python, SQL
Backend:	Spring Boot, REST/gRPC, JSON/Protobuf, WebSockets
Distributed Systems:	Microservices, event driven architecture, Kafka/RabbitMQ/SQS, circuit breaking, idempotency & retries, saga/outbox
LLMs & RAG:	LangChain, Vector Databases (Pinecone, FAISS), Semantic Search, RAG, Agentic AI
Cloud/Infra:	AWS (EC2, ECS/EKS, Lambda, API Gateway, RDS, DynamoDB, S3, IAM, CloudWatch), Docker, Kubernetes, Terraform
Data & ML:	Data pipelines, feature engineering, classic NLP (HMMs, POS tagging), basic LLM orchestration, vector search/RAG
Quality & DevEx:	GitHub Actions/Jenkins, canary releases, feature flags, Open Telemetry/CloudWatch, JUnit 5, Mockito, Wire Mock, Test containers, CI/CD

EDUCATION

Arizona State University — Tempe, AZ, USA
Master of Science in Computer Science — **Jan 2021 - Aug 2022**

Arizona State University — Tempe, AZ, USA
Bachelor of Science in Computer Science — **Aug 2016 - May 2020**